11.Write a case study to identify and analyze the requirements of a Railway Reservation System and prepare positive test scenarios and test cases for the application.

Aim

To identify and analyze the requirements of a Railway Reservation System and prepare positive test scenarios and test cases.

Requirements Analysis

Functional requirements

1. Passenger registration and login.
2. Train search by source, destination, and travel date.
3. Seat availability checking.
4. Passenger information entry.
5. Ticket booking and payment.
6. PNR or booking reference generation.
7. Ticket viewing and downloading.
8. Booking history and cancellation.

Non-functional requirements

- Accurate seat allocation.
- Secure passenger and payment data.
- Reliable booking and payment processing.
- Acceptable response time during high demand.

Positive Test Scenarios

Scenario ID	**Test scenario**	**Expected result**
RPS01	Login with valid credentials	Login successful
RPS02	Search trains using valid journey details	Matching trains displayed
RPS03	Check available seats	Correct availability displayed

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RPS04	Enter valid passenger details	Details accepted
RPS05	Book a ticket when seats are available	Booking successful
RPS06	Complete payment successfully	Payment confirmed
RPS07	Generate booking reference	Unique reference generated
RPS08	View booked ticket	Correct ticket details displayed
RPS09	Download or print ticket	Ticket downloaded successfully
RPS10	View booking history	Previous bookings displayed

Positive Test Cases

Test Case ID: RRS_TC01

1. Test case: Login with valid credentials.
2. Precondition: Passenger has a registered account.
3. Steps to execute:
    1. Open the reservation application.
    2. Enter valid credentials.
    3. Click Login.
4. Test data: Valid username and password.
5. Expected result: Passenger dashboard is displayed.
6. Status: Pass/Fail

Test Case ID: RRS_TC02

1. Test case: Search for trains.
2. Precondition: Train schedules are available.
3. Steps to execute:
    1. Enter source station.
    2. Enter destination station.
    3. Select a valid travel date.

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    4. Click Search.

4. Test data: Chennai to Bengaluru, with a valid future travel date.
5. Expected result: Trains matching the journey details are displayed.
6. Status: Pass/Fail

Test Case ID: RRS_TC03

1. Test case: Check seat availability.
2. Precondition: A matching train is displayed.
3. Steps to execute:
    1. Select a train.
    2. Choose the travel class.
    3. Check seat availability.
4. Test data: Valid train, date, and class.
5. Expected result: Current availability or waitlist status is displayed correctly.
6. Status: Pass/Fail

Test Case ID: RRS_TC04

1. Test case: Enter valid passenger details.
2. Precondition: Passenger has selected a train.
3. Steps to execute:
    1. Open the booking form.
    2. Enter passenger name, age, and other required details.
    3. Continue to payment.
4. Test data: Valid passenger details.
5. Expected result: Details are accepted and the booking process continues.
6. Status: Pass/Fail

Test Case ID: RRS_TC05

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1. Test case: Book a ticket when seats are available.
2. Precondition: A seat is available and passenger details are valid.
3. Steps to execute:
    1. Select a train and class.
    2. Enter passenger details.
    3. Confirm the booking.
    4. Complete the required payment.
4. Test data: Valid journey and passenger details.
5. Expected result: Booking is confirmed and a seat or berth is allocated according to system rules.
6. Status: Pass/Fail

Test Case ID: RRS_TC06

1. Test case: Complete ticket payment.
2. Precondition: Booking is ready for payment.
3. Steps to execute:
    1. Select a valid payment method.
    2. Complete the payment in the test environment.
4. Test data: Successful test payment response.
5. Expected result: Payment is confirmed and the booking status is updated.
6. Status: Pass/Fail

Test Case ID: RRS_TC07

1. Test case: Generate booking reference.
2. Precondition: Booking is successfully confirmed.
3. Steps to execute:
    1. Complete a valid booking.

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    2. Open booking confirmation.

4. Test data: Confirmed booking.
5. Expected result: A unique PNR or booking reference is generated and displayed.
6. Status: Pass/Fail

Test Case ID: RRS_TC08

1. Test case: View booked ticket.
2. Precondition: A confirmed booking exists.
3. Steps to execute:
    1. Open My Bookings.
    2. Select the confirmed ticket.
4. Test data: Valid booking reference.
5. Expected result: Correct passenger, train, date, class, and booking details are displayed.
6. Status: Pass/Fail

Test Case ID: RRS_TC09

1. Test case: Download or print ticket.
2. Precondition: A booking exists.
3. Steps to execute:
    1. Open the ticket details.
    2. Click Download or Print.
4. Test data: Confirmed ticket.
5. Expected result: Ticket is generated with correct booking information.
6. Status: Pass/Fail

Test Case ID: RRS_TC10

1. Test case: View booking history.
2. Precondition: Passenger has previous bookings.

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3. Steps to execute:
    1. Log in.
    2. Open My Bookings.
4. Test data: Registered account with booking records.
5. Expected result: Previous bookings and their statuses are displayed.
6. Status: Pass/Fail

Result

Ten positive test scenarios and test cases were prepared to validate train search, seat availability, passenger information, ticket booking, payment, booking references, and ticket management.

12.Write a case study to identify and analyze the requirements of a Railway Reservation System and prepare negative test scenarios and test cases for the application.

Aim

To identify and analyze the requirements of a Railway Reservation System and prepare negative test scenarios and test cases.

Requirements Analysis

Functional requirements

1. Validate passenger login.
2. Validate source, destination, and travel date.
3. Check seat availability before booking.
4. Validate passenger details.
5. Handle failed payments.
6. Prevent duplicate bookings.
7. Restrict unauthorized access to booking details.
8. Validate ticket cancellation requests.

Non-functional requirements

- Protect passenger data.

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- Maintain accurate seat availability.
- Prevent duplicate charges and bookings.
- Display appropriate error messages.

Negative Test Scenarios

Scenario ID	Test scenario	Expected result
RNS01	Login with incorrect credentials	Login rejected
RNS02	Search without entering source or destination	Validation displayed
RNS03	Select an invalid travel date	Search rejected
RNS04	Attempt booking when no seats are available	Booking rejected or waitlist offered
RNS05	Submit passenger details with missing fields	Validation displayed
RNS06	Enter an invalid passenger age	Invalid age rejected
RNS07	Payment fails during booking	Booking not marked as paid
RNS08	Submit the same booking request repeatedly	Duplicate booking prevented
RNS09	Access another passenger's ticket	Access denied
RNS10	Cancel a nonexistent booking	Cancellation rejected

Negative Test Cases

Test Case ID: RRS_NTC01

1. Test case: Login with incorrect credentials.
2. Precondition: Login page is available.
3. Steps to execute:
    1. Open the reservation application.
    2. Enter an incorrect password.
    3. Click Login.
4. Test data: Valid username and incorrect password.
5. Expected result: Login is rejected.

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6. Status: Pass/Fail

Test Case ID: RRS_NTC02

1. Test case: Search without source or destination.
2. Precondition: Train search page is available.
3. Steps to execute:
    1. Open train search.
    2. Leave the source or destination field empty.
    3. Click Search.
4. Test data: Missing source or destination.
5. Expected result: Required-field validation is displayed.
6. Status: Pass/Fail

Test Case ID: RRS_NTC03

1. Test case: Select an invalid travel date.
2. Precondition: Train search is available.
3. Steps to execute:
    1. Enter valid source and destination.
    2. Select a past travel date.
    3. Click Search.
4. Test data: Past travel date.
5. Expected result: The system rejects the invalid travel date.
6. Status: Pass/Fail

Test Case ID: RRS_NTC04

1. Test case: Attempt booking when seats are unavailable.
2. Precondition: Selected train and class have no bookable seats.
3. Steps to execute:

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    1. Search for a train.
    2. Select a class with no available seats.
    3. Attempt to book.
4. Test data: Fully booked train.
5. Expected result: The system prevents an invalid confirmed booking or offers an eligible waitlist option.
6. Status: Pass/Fail

Test Case ID: RRS_NTC05

1. Test case: Submit passenger details with missing fields.
2. Precondition: Booking form is available.
3. Steps to execute:
    1. Open the passenger details form.
    2. Leave a required field empty.
    3. Click Continue.
4. Test data: Incomplete passenger details.
5. Expected result: The system rejects the incomplete form.
6. Status: Pass/Fail

Test Case ID: RRS_NTC06

1. Test case: Enter an invalid passenger age.
2. Precondition: Passenger details form is available.
3. Steps to execute:
    1. Enter an invalid age, such as a negative value.
    2. Submit the passenger details.
4. Test data: Age -5.
5. Expected result: Invalid age is rejected according to the application's validation rules.

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6. Status: Pass/Fail

Test Case ID: RRS_NTC07

1. Test case: Handle payment failure.
2. Precondition: A booking is ready for payment.
3. Steps to execute:
    1. Proceed to payment.
    2. Simulate a failed payment in the test environment.
    3. Check booking and payment status.
4. Test data: Failed payment response.
5. Expected result: The booking must not be marked as paid; its status must reflect the actual transaction outcome.
6. Status: Pass/Fail

Test Case ID: RRS_NTC08

1. Test case: Prevent duplicate bookings.
2. Precondition: Booking request is ready to submit.
3. Steps to execute:
    1. Submit the booking request.
    2. Repeat the same request or retry after a timeout.
    3. Check the booking history.
4. Test data: Same booking request submitted twice.
5. Expected result: Duplicate booking creation is prevented for the same transaction.
6. Status: Pass/Fail

Test Case ID: RRS_NTC09

1. Test case: Access another passenger's ticket.
2. Precondition: Passenger is logged in.

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3. Steps to execute:
    1. Attempt to open a ticket belonging to another passenger.
    2. Submit the unauthorized request.
4. Test data: Unauthorized booking reference.
5. Expected result: Access is denied.
6. Status: Pass/Fail

Test Case ID: RRS_NTC10

1. Test case: Cancel a nonexistent booking.
2. Precondition: Cancellation function is available.
3. Steps to execute:
    1. Open the cancellation page.
    2. Enter a nonexistent booking reference.
    3. Submit the request.
4. Test data: Invalid booking reference.
5. Expected result: Cancellation is rejected and an appropriate message is displayed.
6. Status: Pass/Fail

Result

Ten negative test scenarios and test cases were prepared to validate journey search, passenger details, seat availability, payment handling, duplicate booking prevention, and ticket access control.

13.Write a case study to identify and analyze the requirements of an Online Examination System and prepare positive test scenarios and test cases for the application.

Aim

To identify and analyze the requirements of an Online Examination System and prepare positive test scenarios and test cases.

Requirements Analysis

Functional requirements

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1. Student and administrator login.
2. Examination scheduling.
3. Displaying instructions and questions.
4. Saving student responses.
5. Navigating between questions.
6. Submitting examinations.
7. Automatic evaluation of objective questions.
8. Publishing results.
9. Managing questions and exam settings.

Non-functional requirements

- Maintain examination integrity.
- Save answers reliably.
- Handle concurrent students.
- Protect examination data.
- Provide a user-friendly interface.

Positive Test Scenarios

Scenario ID	**Test scenario**	**Expected result**
OES_PS01	Student login with valid credentials	Login successful
OES_PS02	View scheduled examinations	Correct exams displayed
OES_PS03	Open an examination during the permitted period	Exam starts successfully
OES_PS04	Select an answer	Answer is recorded
OES_PS05	Navigate between questions	Questions displayed correctly
OES_PS06	Save answers and resume when permitted	Saved answers restored
OES_PS07	Submit examination before the deadline	Submission successful

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OES_PS08	Calculate objective-question score	Correct score calculated
OES_PS09	View published results	Correct results displayed
OES_PS10	Faculty creates an examination	Exam saved successfully

Positive Test Cases

Test Case ID: OES_TC01

1. Test case: Student login with valid credentials.
2. Precondition: Student has a registered account.
3. Steps to execute:
    1. Open the examination portal.
    2. Enter valid credentials.
    3. Click Login.
4. Test data: Valid student username and password.
5. Expected result: Student dashboard is displayed.
6. Status: Pass/Fail

Test Case ID: OES_TC02

1. Test case: View scheduled examinations.
2. Precondition: Student is logged in and examinations are scheduled.
3. Steps to execute:
    1. Open My Examinations.
    2. View the examination list.
4. Test data: Student account with scheduled exams.
5. Expected result: Correct exam names, dates, and timings are displayed.
6. Status: Pass/Fail

Test Case ID: OES_TC03

1. Test case: Start an examination during the permitted period.

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2. Precondition: Student is eligible and the examination is open.
3. Steps to execute:
    1. Open My Examinations.
    2. Select the scheduled exam.
    3. Read the instructions.
    4. Click Start Exam.
4. Test data: Valid examination and eligible student.
5. Expected result: The examination begins and questions are displayed.
6. Status: Pass/Fail

Test Case ID: OES_TC04

1. Test case: Select an answer.
2. Precondition: Examination is in progress.
3. Steps to execute:
    1. Open a multiple-choice question.
    2. Select an answer.
    3. Move to another question.
4. Test data: Valid answer option.
5. Expected result: The selected answer is saved according to the application's response-saving rules.
6. Status: Pass/Fail

Test Case ID: OES_TC05

1. Test case: Navigate between questions.
2. Precondition: Examination contains multiple questions.
3. Steps to execute:
    1. Open the exam.

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2. Click Next Question.
3. Click Previous Question.

4. Test data: Examination with multiple questions.
5. Expected result: Correct questions are displayed and saved responses are retained.
6. Status: Pass/Fail

Test Case ID: OES_TC06

1. Test case: Save and restore answers.
2. Precondition: The examination supports saving and resuming.
3. Steps to execute:
    1. Answer several questions.
    2. Save the responses.
    3. Resume the exam using the permitted workflow.
4. Test data: Exam with saved responses.
5. Expected result: Previously saved answers are restored correctly.
6. Status: Pass/Fail

Test Case ID: OES_TC07

1. Test case: Submit examination before the deadline.
2. Precondition: Exam is active.
3. Steps to execute:
    1. Answer the questions.
    2. Click Submit Exam.
    3. Confirm submission if required.
4. Test data: Valid examination attempt.
5. Expected result: Responses are submitted, and a confirmation is displayed.
6. Status: Pass/Fail

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Test Case ID: OES_TC08

1. Test case: Evaluate objective questions.
2. Precondition: An answer key is configured.
3. Steps to execute:
    1. Complete a test examination.
    2. Submit responses.
    3. Run the evaluation.
4. Test data: Questions with known correct answers.
5. Expected result: Scores are calculated according to the configured marking scheme.
6. Status: Pass/Fail

Test Case ID: OES_TC09

1. Test case: View published examination results.
2. Precondition: Results have been published.
3. Steps to execute:
    1. Log in as a student.
    2. Open Results.
    3. Select the examination.
4. Test data: Student account with published results.
5. Expected result: Correct marks and result status are displayed.
6. Status: Pass/Fail

Test Case ID: OES_TC10

1. Test case: Create an examination as faculty.
2. Precondition: Faculty has examination-management permission.
3. Steps to execute:
    1. Log in as faculty.

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2. Open Exam Management.
3. Enter exam name, schedule, duration, and questions.
4. Save the examination.

4. Test data: Valid examination details.
5. Expected result: Examination is created and saved successfully.
6. Status: Pass/Fail

Result

Ten positive test scenarios and test cases were prepared to verify login, examination scheduling, question navigation, answer saving, submission, evaluation, result viewing, and exam creation.

14.Write a case study to identify and analyze the requirements of an Online Examination System and prepare negative test scenarios and test cases for the application.

Aim

To identify and analyze the requirements of an Online Examination System and prepare negative test scenarios and test cases.

Requirements Analysis

Functional requirements

1. Validate student and faculty credentials.
2. Restrict access to scheduled examinations.
3. Validate answer submissions.
4. Prevent unauthorized examination attempts.
5. Enforce examination time limits.
6. Prevent duplicate submissions.
7. Restrict access to unpublished results and answer keys.
8. Validate exam creation and scheduling.

Non-functional requirements

- Protect student and examination data.

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- Maintain reliable response saving.
- Prevent unauthorized access.
- Handle interruptions without corrupting submissions.

Negative Test Scenarios

Scenario ID	Test scenario	Expected result
OES_NS01	Login with incorrect credentials	Login rejected
OES_NS02	Open an exam before its start time	Access denied
OES_NS03	Open an exam after the deadline	Access denied
OES_NS04	Start an exam without eligibility	Access denied
OES_NS05	Submit an answer in an invalid format	Input rejected
OES_NS06	Submit an examination twice	Duplicate submission prevented
OES_NS07	Submit after the time limit	Submission handled according to exam rules
OES_NS08	Access another student's results	Access denied
OES_NS09	View unpublished results	Access denied
OES_NS10	Create an exam with missing required details	Validation displayed

Negative Test Cases

Test Case ID: OES_NTC01

1. Test case: Login with incorrect credentials.
2. Precondition: Login page is available.
3. Steps to execute:
    1. Enter an invalid username or password.
    2. Click Login.
4. Test data: Incorrect credentials.

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5. Expected result: Login is rejected.
6. Status: Pass/Fail

Test Case ID: OES_NTC02

1. Test case: Open an examination before its start time.
2. Precondition: Examination is scheduled for a future time.
3. Steps to execute:
    1. Log in as an eligible student.
    2. Open the scheduled examination before its start time.
4. Test data: Future examination schedule.
5. Expected result: The system prevents the exam from starting early.
6. Status: Pass/Fail

Test Case ID: OES_NTC03

1. Test case: Open an examination after the deadline.
2. Precondition: Examination window has closed.
3. Steps to execute:
    1. Log in as a student.
    2. Attempt to start the expired examination.
4. Test data: Examination past its closing time.
5. Expected result: The system denies access to the exam.
6. Status: Pass/Fail

Test Case ID: OES_NTC04

1. Test case: Start an exam without eligibility.
2. Precondition: Student is not registered or authorized for the exam.
3. Steps to execute:
    1. Log in with the student account.

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    2. Attempt to open an examination for which the student is not eligible.
4. Test data: Unauthorized student account.
5. Expected result: The system denies access.
6. Status: Pass/Fail

Test Case ID: OES_NTC05

1. Test case: Submit an answer in an invalid format.
2. Precondition: Examination is in progress and an answer field has format restrictions.
3. Steps to execute:
    1. Open a question requiring a numeric response.
    2. Enter invalid text.
    3. Attempt to save the answer.
4. Test data: ABC for a numeric-only field.
5. Expected result: Invalid input is rejected and an appropriate validation message is displayed.
6. Status: Pass/Fail

Test Case ID: OES_NTC06

1. Test case: Prevent duplicate examination submission.
2. Precondition: Student has an active exam attempt.
3. Steps to execute:
    1. Submit the examination.
    2. Repeat the submission request.
    3. Check the submission record.
4. Test data: Same exam attempt submitted twice.
5. Expected result: Only one valid submission is recorded for the attempt.
6. Status: Pass/Fail

Test Case ID: OES_NTC07

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1. Test case: Handle submission after the time limit.
2. Precondition: Examination duration has expired.
3. Steps to execute:
    1. Start an exam.
    2. Allow the configured duration to expire.
    3. Attempt to submit additional answers.
4. Test data: Expired exam attempt.
5. Expected result: The system enforces the configured deadline and handles saved answers according to exam rules.
6. Status: Pass/Fail

Test Case ID: OES_NTC08

1. Test case: Access another student's results.
2. Precondition: Student is logged in.
3. Steps to execute:
    1. Attempt to access another student's result using an unauthorized identifier.
    2. Submit the request.
4. Test data: Unauthorized student ID.
5. Expected result: Access is denied and student marks are not disclosed.
6. Status: Pass/Fail

Test Case ID: OES_NTC09

1. Test case: View unpublished examination results.
2. Precondition: Results have not been released.
3. Steps to execute:
    1. Log in as a student.
    2. Open the Results section.

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3. Attempt to view unreleased results.

4. Test data: Unpublished examination.
5. Expected result: The system does not disclose results before publication.
6. Status: Pass/Fail

Test Case ID: OES_NTC10

1. Test case: Create an examination with missing required details.
2. Precondition: Faculty has exam-creation permission.
3. Steps to execute:
    1. Open Exam Management.
    2. Leave the exam name or duration empty.
    3. Click Save.
4. Test data: Incomplete exam details.
5. Expected result: The system rejects the incomplete form and highlights required fields.
6. Status: Pass/Fail

Result

Ten negative test scenarios and test cases were prepared to validate authentication, exam scheduling, eligibility, input validation, submission controls, result confidentiality, and exam configuration.

15.Develop a comprehensive Test Plan document for an Inventory Management System, covering test scope, test design techniques, test execution strategy, and test completion criteria.

Aim

To develop a comprehensive Test Plan document for an Inventory Management System, covering test scope, test design techniques, execution strategy, and test completion criteria.

1. Introduction

An Inventory Management System is a software application used to manage products, stock quantities, suppliers, purchases, sales, and stock movements. The test plan defines how the system will be evaluated to ensure accurate inventory tracking and reliable operations.

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2. Objectives

- Verify accurate stock calculations.
- Validate product and supplier management.
- Test stock-in and stock-out transactions.
- Ensure purchase and sales records are correct.
- Identify and document software defects.
- Verify role-based access and data integrity.

3. Test Scope

In scope

1. User login and logout.
2. Product creation, update, and deletion.
3. Category management.
4. Supplier management.
5. Stock-in and stock-out operations.
6. Purchase and sales records.
7. Low-stock alerts.
8. Inventory reports.
9. Input validation and access control.

Out of scope

1. Testing physical barcode scanner hardware.
2. Testing external supplier systems.
3. Testing unrelated accounting applications.

4. Testing Strategies and Methodologies

Testing type	**Purpose**

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Functional testing	Verify inventory operations
Positive testing	Verify valid stock transactions
Negative testing	Verify rejection of invalid quantities and product data
Integration testing	Verify interactions between products, suppliers, and stock transactions
System testing	Test the complete inventory application
Regression testing	Verify that updates do not break existing functions
Performance testing	Check performance under expected transaction volumes
Security testing	Verify access restrictions and data protection
User acceptance testing	Confirm the system meets store or warehouse requirements

5. Test Design Techniques

1. Equivalence partitioning: Divide product quantities into valid and invalid groups.
2. Boundary value analysis: Test quantities at zero, one, maximum stock, and permitted limits.
3. Decision table testing: Verify stock availability, order quantity, and user permissions together.
4. State transition testing: Check product states such as in stock, low stock, and out of stock.
5. Error guessing: Test duplicate product IDs, missing supplier details, and invalid stock adjustments.

6. Sample Test Scenarios

Test case ID	**Test scenario**	**Expected result**
IMS_TC01	Login with valid credentials	Dashboard displayed
IMS_TC02	Add a new product with valid details	Product added
IMS_TC03	Update existing product details	Changes saved
IMS_TC04	Record stock received from a supplier	Stock quantity increased
IMS_TC05	Record a valid stock-out transaction	Stock quantity decreased
IMS_TC06	Generate an inventory report	Accurate report displayed

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IMS_TC07	Enter a negative stock quantity	Invalid quantity rejected
IMS_TC08	Add a duplicate product ID	Duplicate rejected
IMS_TC09	Issue more stock than available	Transaction rejected
IMS_TC10	Access administrator functions without permission	Access denied

7. Test Environment

- Hardware: Computer or laptop with network access.
- Software: Inventory application and compatible web browser.
- Database: Sample products, suppliers, stock levels, and transaction records.
- Tools: Test-case spreadsheet, defect-tracking sheet, and optional automation tools.

8. Test Execution Strategy

1. Review and analyze inventory requirements.
2. Prepare test scenarios and test cases.
3. Configure the test environment and sample data.
4. Execute positive and negative tests.
5. Record expected and actual results.
6. Report defects with IDs, severity, priority, and reproduction steps.
7. Retest fixed defects.
8. Execute regression tests.
9. Prepare the final test summary.

9. Entry Criteria

- Requirements are reviewed and approved.
- The test application is available.
- Test data and test cases are prepared.
- Required user accounts and permissions are configured.

10. Exit Criteria

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- All planned critical test cases have been executed.
- Critical and high-severity defects are resolved or formally accepted.
- Fixed defects have been retested.
- Regression testing is completed.
- Test results and outstanding defects are documented.

11. Deliverables

1. Test Plan document.
2. Test scenarios and test cases.
3. Test execution report.
4. Defect reports.
5. Regression test results.
6. Final Test Summary Report.

12. Roles and Responsibilities

Role	**Responsibility**
Test Lead	Plan and monitor testing
Tester	Design and execute test cases
Developer	Fix defects
Inventory Manager	Validate business operations during acceptance testing

Result

A comprehensive Test Plan was developed for the Inventory Management System, covering test scope, testing methodologies, test design techniques, execution strategy, entry and exit criteria, and test deliverables.

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